

CYBER SECURITY LAB EXPERIMENT

Title: Privacy Audit of Popular Apps (Desktop **WhatsApp**) and Websites (**Facebook**) & Data Breach Case Study Analysis

Aim

To analyze privacy, network behavior, and security risks of applications and websites using Kali Linux tools.

Tools Required

- **Wireshark**
- **Burp Suite**
- Kali Linux (VirtualBox)
- Desktop WhatsApp (Nativefier)
- Web Browser (Firefox)

Step 1: Install Node.js & npm

```
sudo apt update
```

The screenshot displays a Kali Linux desktop environment. The top panel shows the system menu with icons for applications, internet, and system status. The terminal window is open, showing the following content:

```

kali@kali: ~
Session Actions Edit View Help

Loaded: loaded (/usr/lib/systemd/system/snapd.service; enabled; preset:~>
Active: inactive (dead) since Wed 2026-04-01 00:57:45 EDT; 1min 0s ago
Duration: 8:35:11
Invocation: b267458db28aa51894ic1150bbbc9875
TriggeredBy: ~ snapd.socket
Docs: man:Snapd.service.5
Process: 9697 /usr/lib/systemd/systemd --no-pager --no-notify --no-acti>
Main PID: 9697 (code=exited, status=42)
Mem peak: 18.2M
CPU: 1.575s

Apr 01 00:57:39 kali snapd[9697]: snapmgr.go:1659: performing periodic snap >
Apr 01 00:57:39 kali snapd[9697]: snapmgr.go:1669: cannot clean store downlo>
Apr 01 00:57:42 kali snapd[9697]: standby.go:181: standby conditions met, in>
Apr 01 00:57:42 kali snapd[9697]: daemon.go:556: gracefully waiting for runs>
Apr 01 00:57:42 kali snapd[9697]: daemon.go:558: done waiting for running ho>
Apr 01 00:57:45 kali snapd[9697]: standby.go:121: standby monitoring stop re>
Apr 01 00:57:45 kali snapd[9697]: overlord.go:543: Released state lock file >
Apr 01 00:57:45 kali snapd[9697]: daemon stop requested to wait for socket a>
Apr 01 00:57:45 kali systemd[1]: snapd.service: Deactivated successfully.
Apr 01 00:57:45 kali systemd[1]: snapd.service: Consumed 1.575s CPU time ove>

kali@kali: ~$
kali@kali: ~$ sudo snap install whatsapp-for-linux
error: snap "whatsapp-for-linux" not found

kali@kali: ~$
kali@kali: ~$ sudo apt update
Hit:1 http://http.kali.org/kali kali-rolling InRelease
1814 packages can be upgraded. Run 'apt list --upgradable' to see them.

kali@kali: ~$

```

The desktop background is a dark-themed Kali Linux logo. The taskbar at the bottom contains icons for the Dash, Home, and various applications. The system tray in the bottom right corner shows the date and time as 10:32 on 01-04-2026.

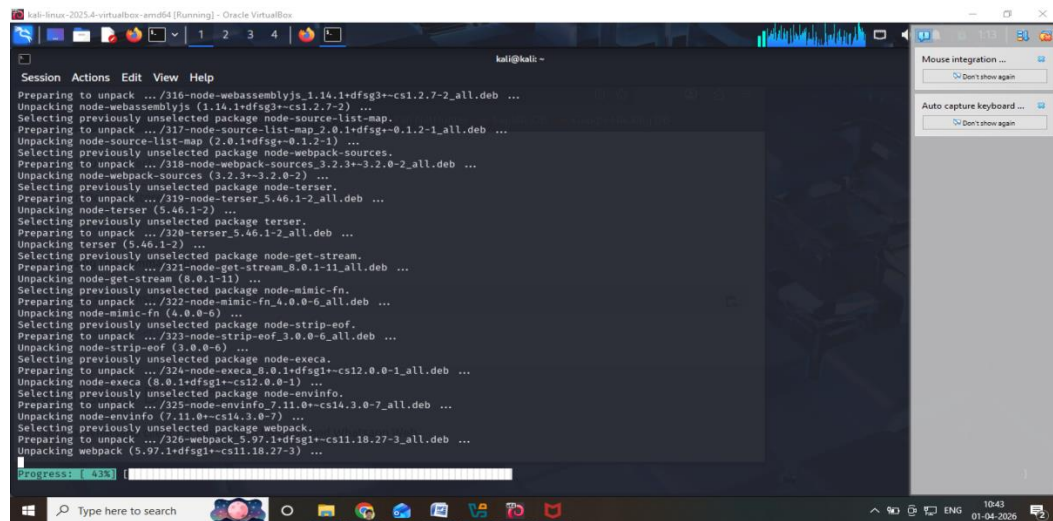
```
sudo apt install nodejs npm -y
```

```
kali-linux-2025.4-virtualbox-amd64 [Running] - Oracle VM VirtualBox
kali@kali: ~
Session Actions Edit View Help
(kali@kali)~$ sudo apt update
Hit:1 http://http.kali.org/kali kali-rolling InRelease
1814 packages can be upgraded. Run 'apt list --upgradable' to see them.

(kali@kali)~$ sudo apt install nodejs npm -y
Upgrading:
libssl3t64 node-acorn nodejs
  in Desktop store

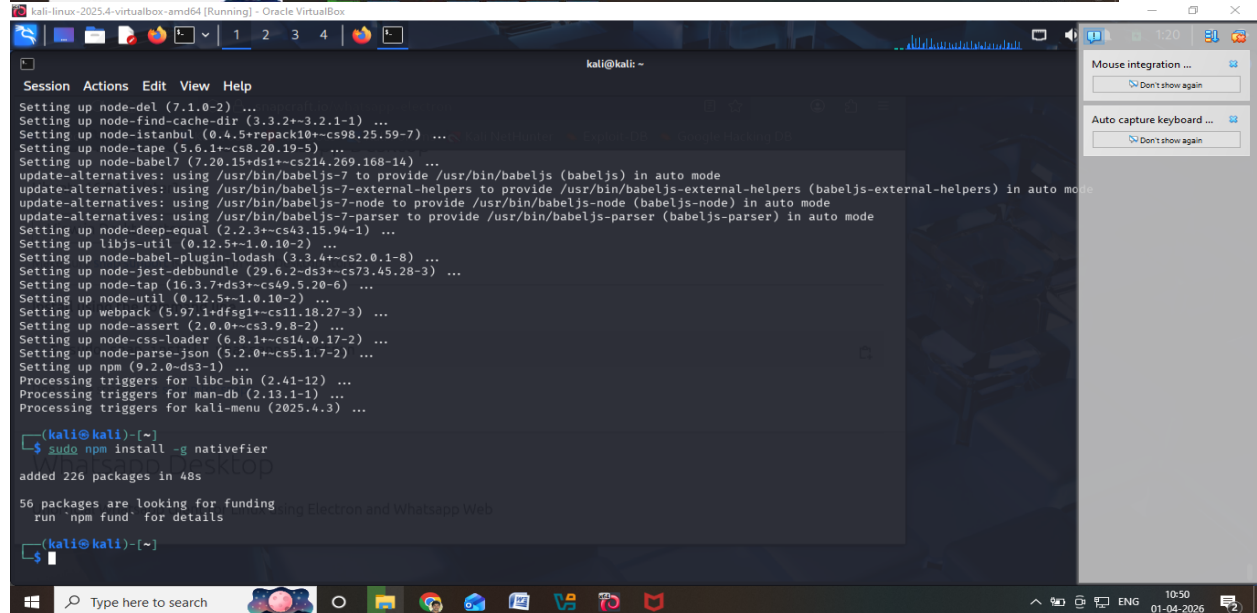
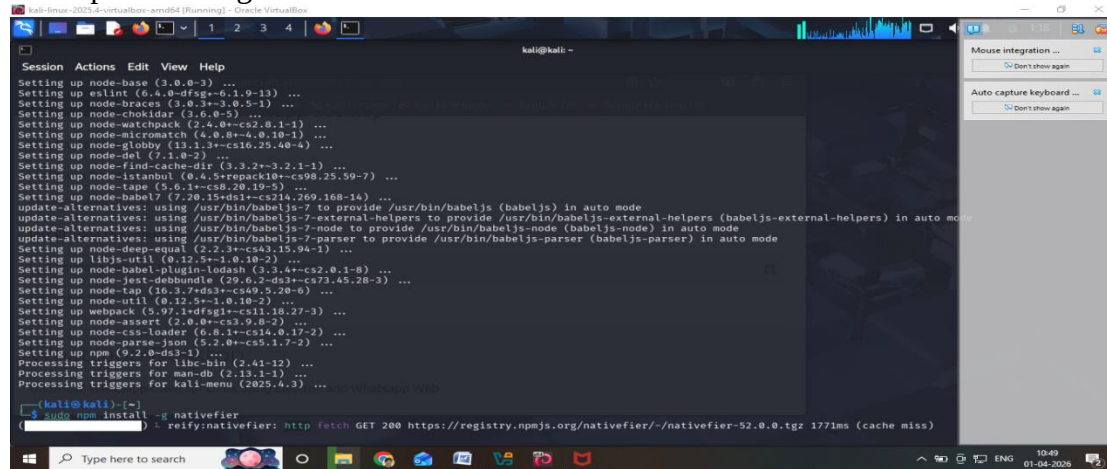
Installing:
npm
  in Desktop store

Installing dependencies:
eslint                  node-delegates          node-jest-worker        node-regenerator-transform
gyp                    node-depd               node-js-tokens          node-regexp
handlebars             node-diff               node-js-yaml            node-regexp-core
libada-url0-3         node-doctrine           node-jscast             node-regjsen
libicu78              node-electron-to-chromium node-json-buffer        node-regparser
libjs-events          node-encoding           node-json-parse-better-errors node-repeat-string
libjs-inherits        node-enhanced-resolve   node-json-schema        node-require-directory
libjs-prettify        node-envinfo            node-json-schema-traverse node-require-from-string
libjs-regenerate      node-err-code           node-json-stable-stringify node-resolve
libjs-source-map      node-errno              node-json5              node-resolve-cwd
libjs-sprintf-js      node-error-ex           node-jsonify            node-resolve-from
libjs-uglify          node-es-abstract        node-jsonparse          node-restore-cursor
libnode-dev           node-es-module-lexer    node-kind-of            node-resumer
libnode127            node-es6-error          node-levn               node-retry
libsimdjson30        node-escape-string-regexp node-loader-runner      node-rimraf
libsimdutf31         node-escodegen          node-locate-path        node-run-queue
libssl-dev            node-eslint-scope       node-lodash             node-safe-buffer
node-abbrev           node-eslint-utils       node-lodash-packages    node-schema-utils
node-agent-base      node-eslint-visitor-keys node-log-driver          node-serialize-javascript
node-ajv              node-esprex             node-log-driver          node-set-blocking
```



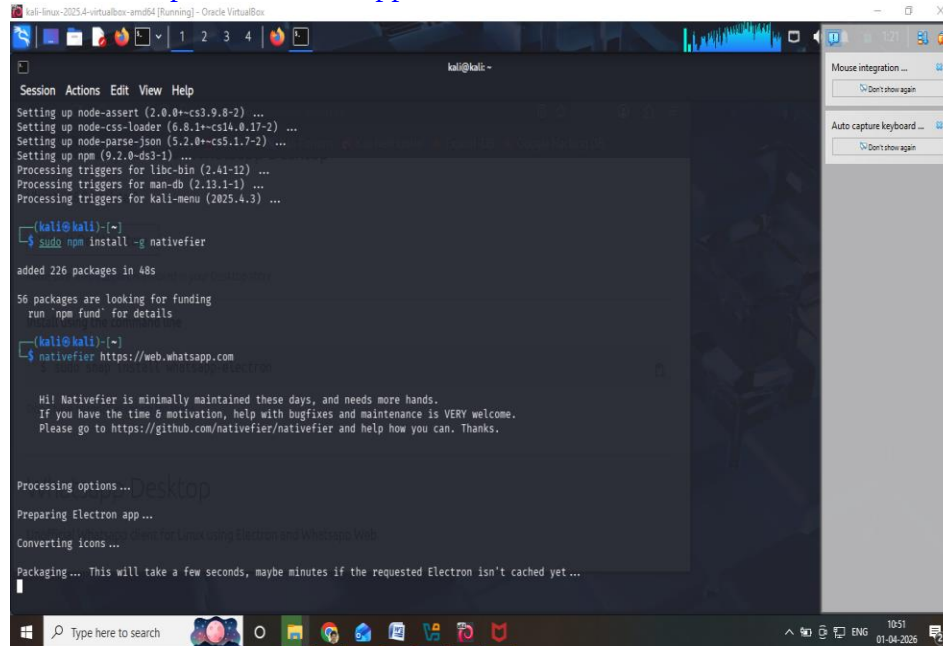
Step 2: Install Nativiefier

sudo npm install -g nativiefier



Step 3: Create WhatsApp Desktop App

nativefier <https://web.whatsapp.com>



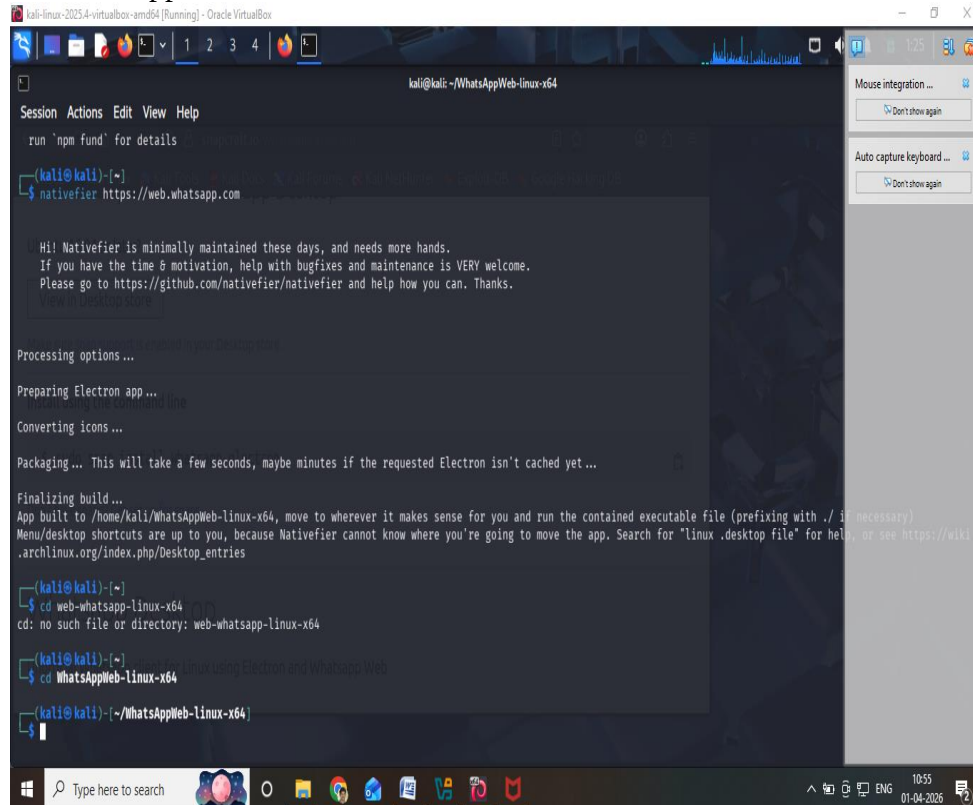
```
kali@kali: ~  
$ sudo npm install -g nativefier  
added 226 packages in 48s  
56 packages are looking for funding  
  run `npm fund` for details  
  
$ nativefier https://web.whatsapp.com  
Processing options... Desktop  
Preparing Electron app...  
Converting icons...  
Packaging... This will take a few seconds, maybe minutes if the requested Electron isn't cached yet...
```

It will create a folder like:

web-whatsapp-linux-x64

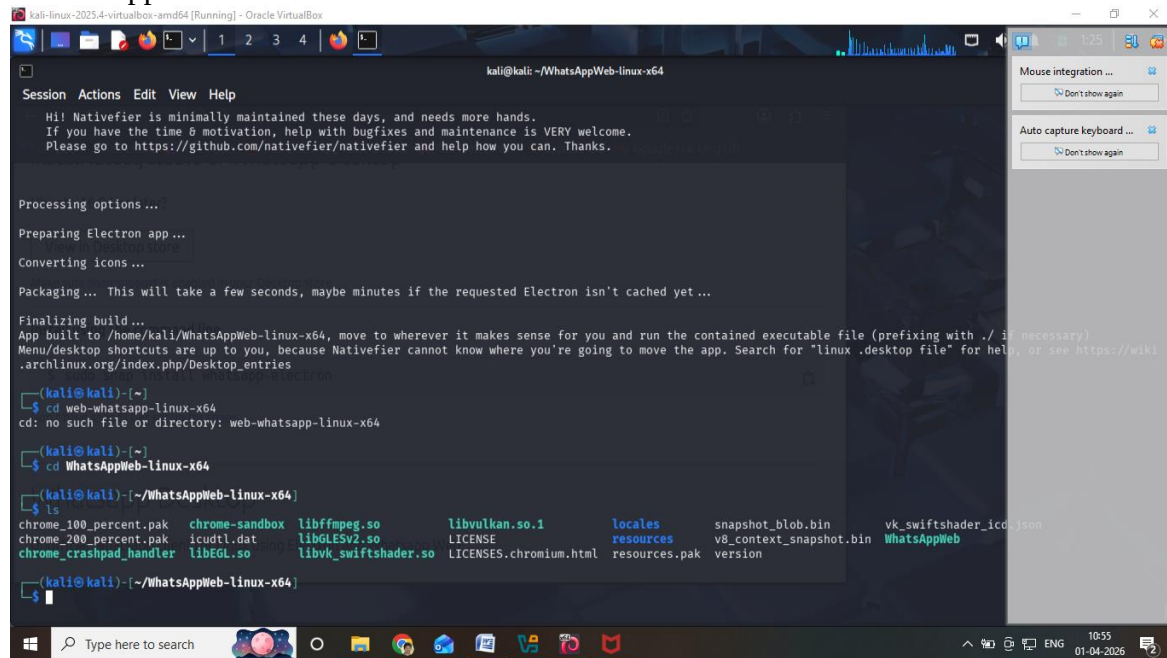
Step 4: Run WhatsApp

cd WhatsAppWeb-linux-x64



```
kali@kali: ~/WhatsAppWeb-linux-x64  
$ nativefier https://web.whatsapp.com  
Processing options... Desktop  
Preparing Electron app...  
Converting icons...  
Packaging... This will take a few seconds, maybe minutes if the requested Electron isn't cached yet...  
Finalizing build...  
App built to /home/kali/WhatsAppWeb-linux-x64, move to wherever it makes sense for you and run the contained executable file (prefixing with ./ if necessary)  
Menu/desktop shortcuts are up to you, because Nativefier cannot know where you're going to move the app. Search for "linux .desktop file" for help, or see https://wiki.archlinux.org/index.php/Desktop_entries  
  
$ cd web-whatsapp-linux-x64  
cd: no such file or directory: web-whatsapp-linux-x64  
  
$ cd WhatsAppWeb-linux-x64  
$
```

./WhatsAppWeb



```
kali@kali: ~/WhatsAppWeb-linux-x64
Session Actions Edit View Help
Hi! Nativefier is minimally maintained these days, and needs more hands.
If you have the time & motivation, help with bugfixes and maintenance is VERY welcome.
Please go to https://github.com/nativefier/nativefier and help how you can. Thanks.

Processing options...
Preparing Electron app...
Converting icons...
Packaging... This will take a few seconds, maybe minutes if the requested Electron isn't cached yet...
Finalizing build...
App built to /home/kali/WhatsAppWeb-linux-x64, move to wherever it makes sense for you and run the contained executable file (prefixed with ./ if necessary)
Menu/desktop shortcuts are up to you, because Nativefier cannot know where you're going to move the app. Search for "linux .desktop file" for help, or see https://wiki.archlinux.org/index.php/Desktop_entries

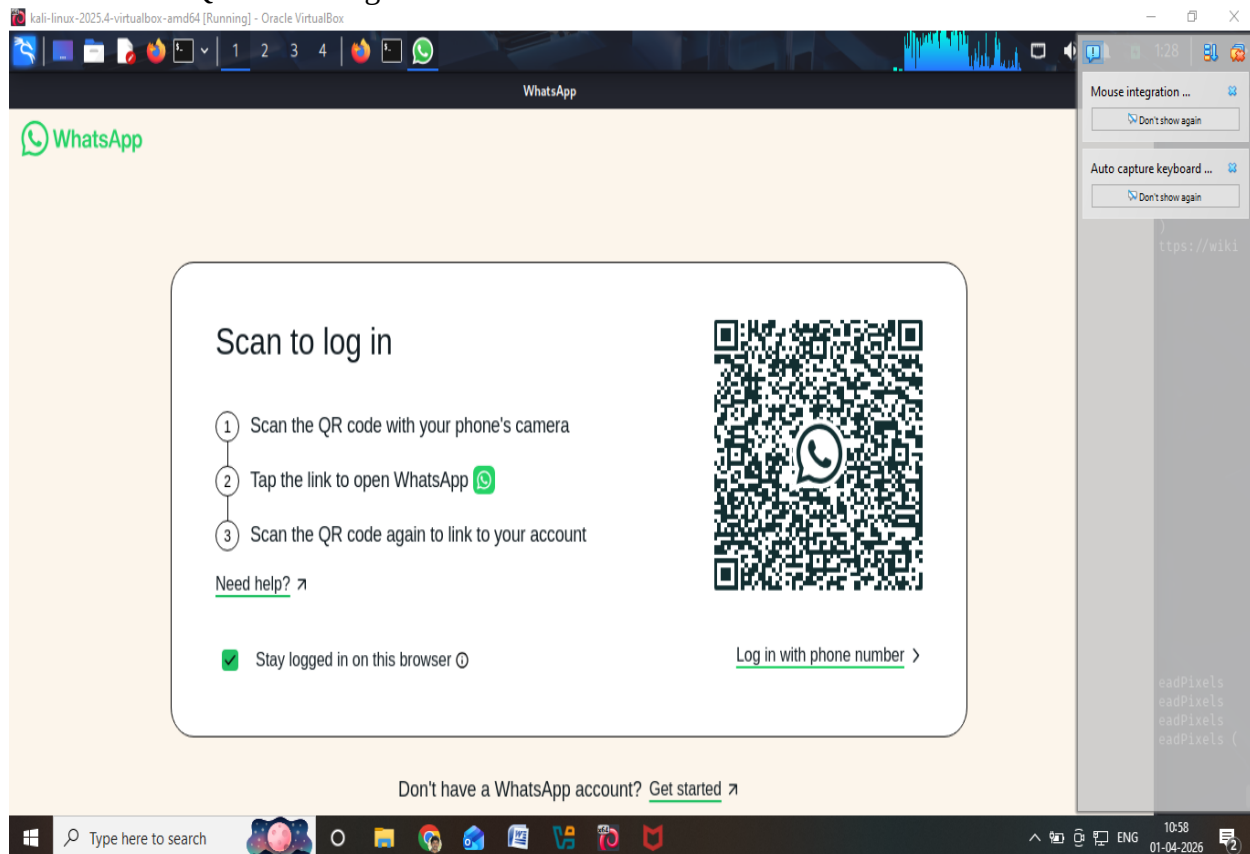
(kali@kali)~$ cd web-whatsapp-linux-x64
cd: no such file or directory: web-whatsapp-linux-x64

(kali@kali)~$ cd WhatsAppWeb-linux-x64

(kali@kali)~/WhatsAppWeb-linux-x64$ ls
chrome_100_percent.pak  chrome-sandbox  libffmpeg.so  libvulkan.so.1  locales  snapshot_blob.bin  vk_swiftshader_icd.json
chrome_200_percent.pak  icudtl.dat      libGLESv2.so  LICENSE         resources  v8_context_snapshot.bin  WhatsAppWeb
chrome_crashpad_handler  libEGL.so       libvk_swiftshader.so  LICENSES.chromium.html  resources.pak  version
```

✔ Step 5: Login

- Scan QR code using mobile



- Done

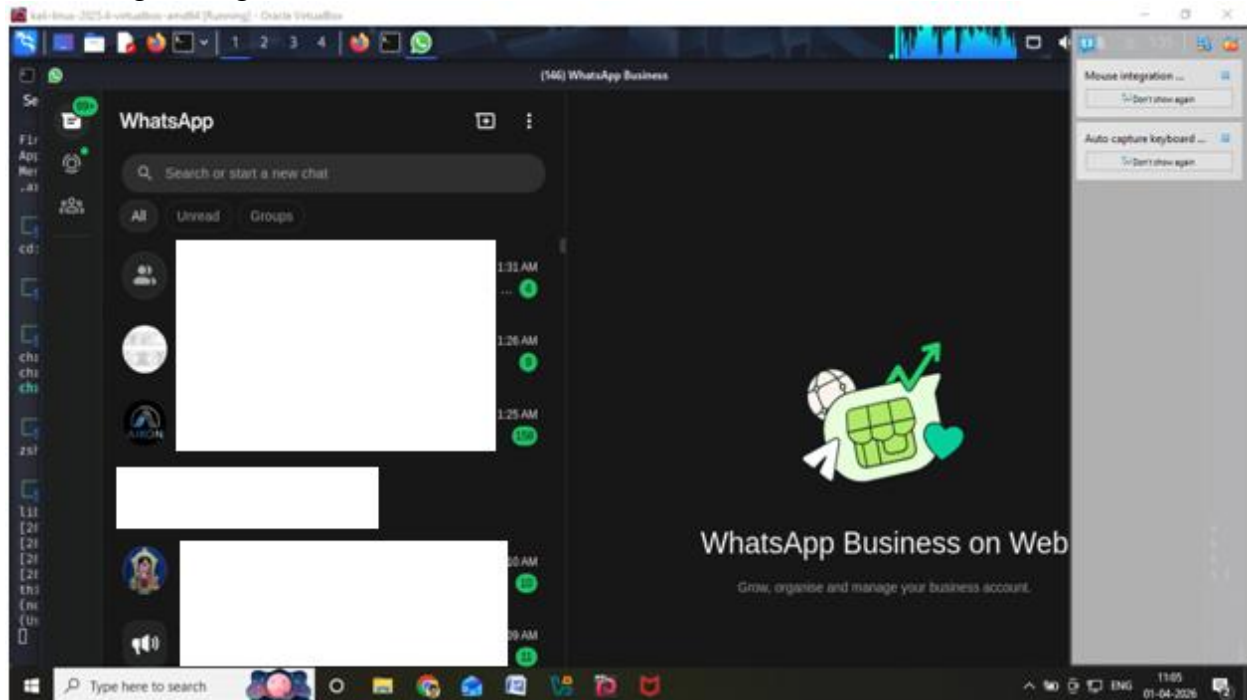
PART 1: Privacy Audit of Desktop WhatsApp

Step 1: Launch WhatsApp Desktop

cd WhatsAppWeb-linux-x64

./WhatsAppWeb

- QR Code appears
- Login using mobile → Linked Devices

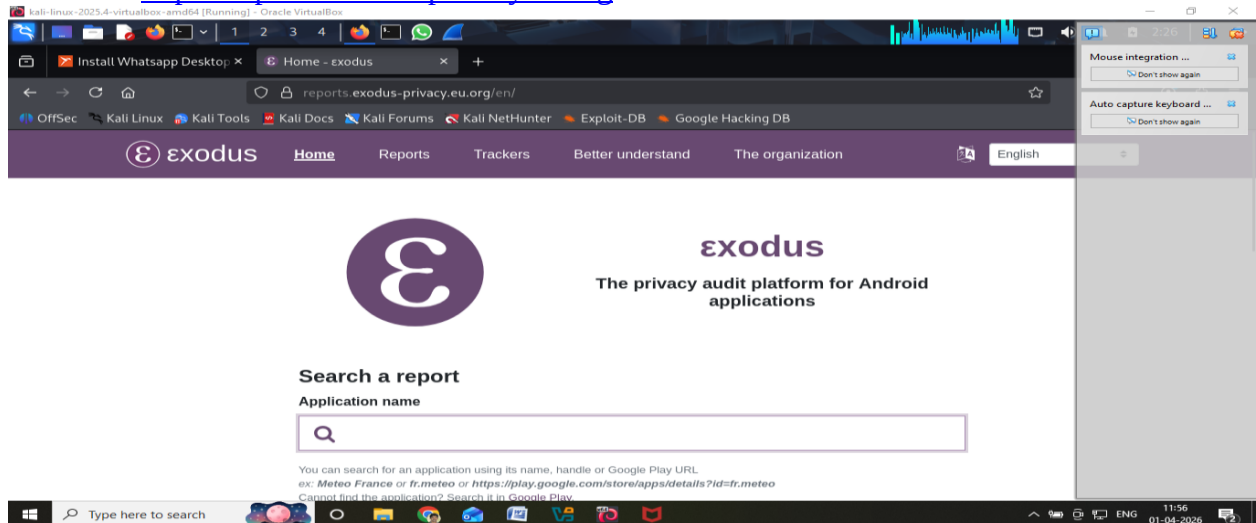


Step 1: Install WhatsApp APK (Optional in Emulator)

- Download APK from trusted source or use your mobile

Step 2: Analyze Trackers using Exodus Privacy

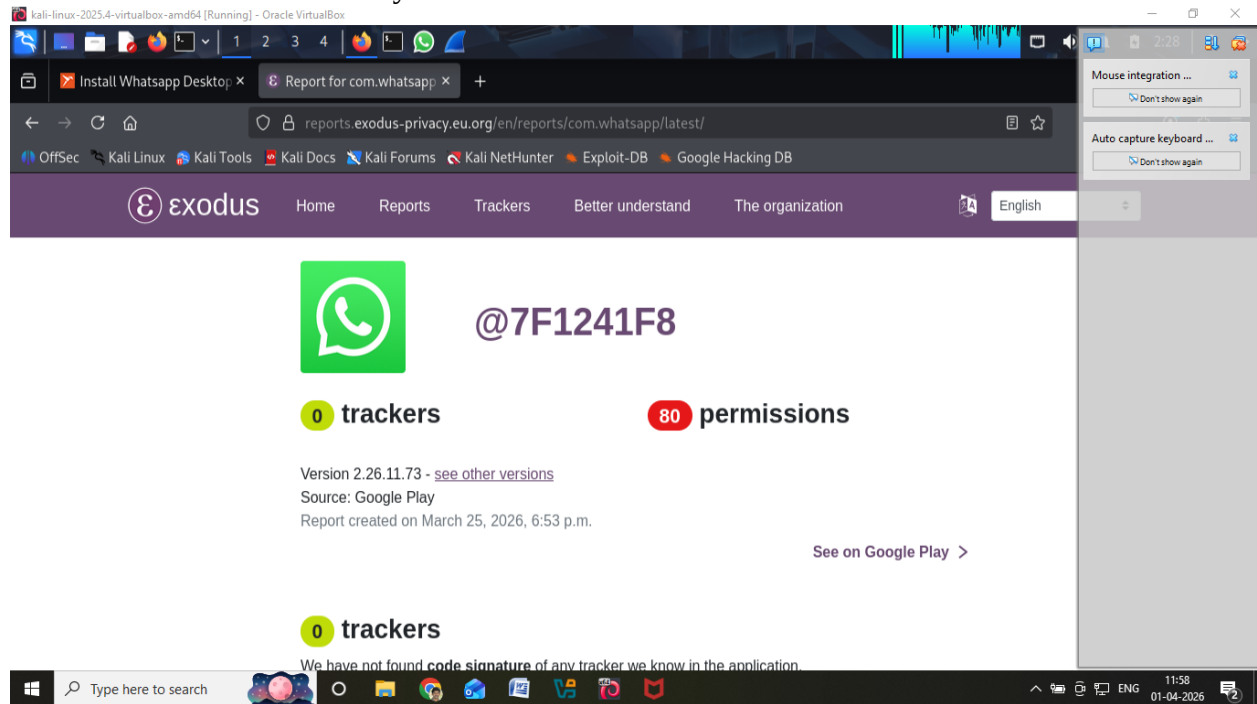
1. Open browser in Kali
2. Go to: <https://reports.exodus-privacy.eu.org>



3. Search for **WhatsApp**

Observation:

- Detects trackers like:
 - Google Analytics
 - Crashlytics
 - Facebook Analytics

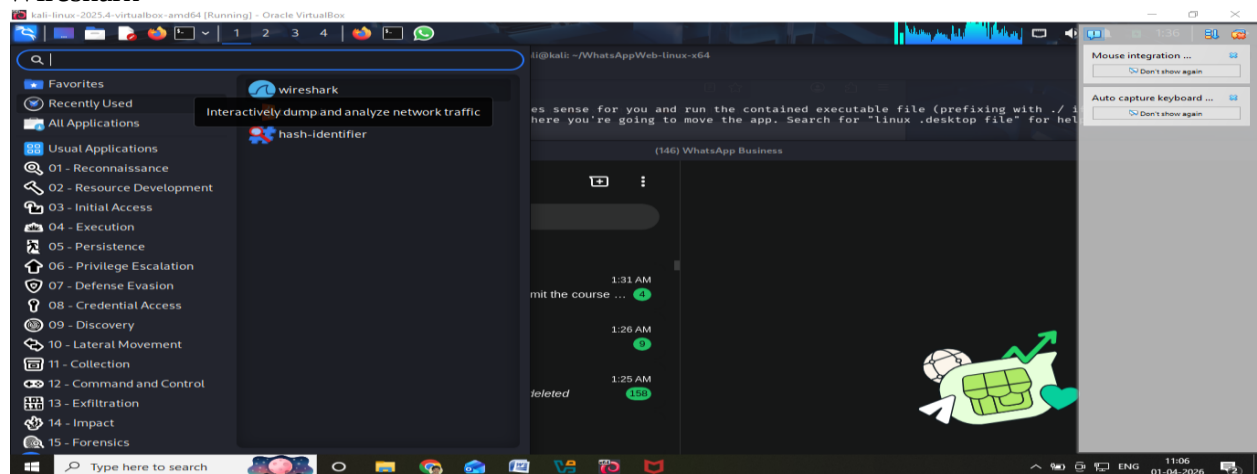


Record in lab:

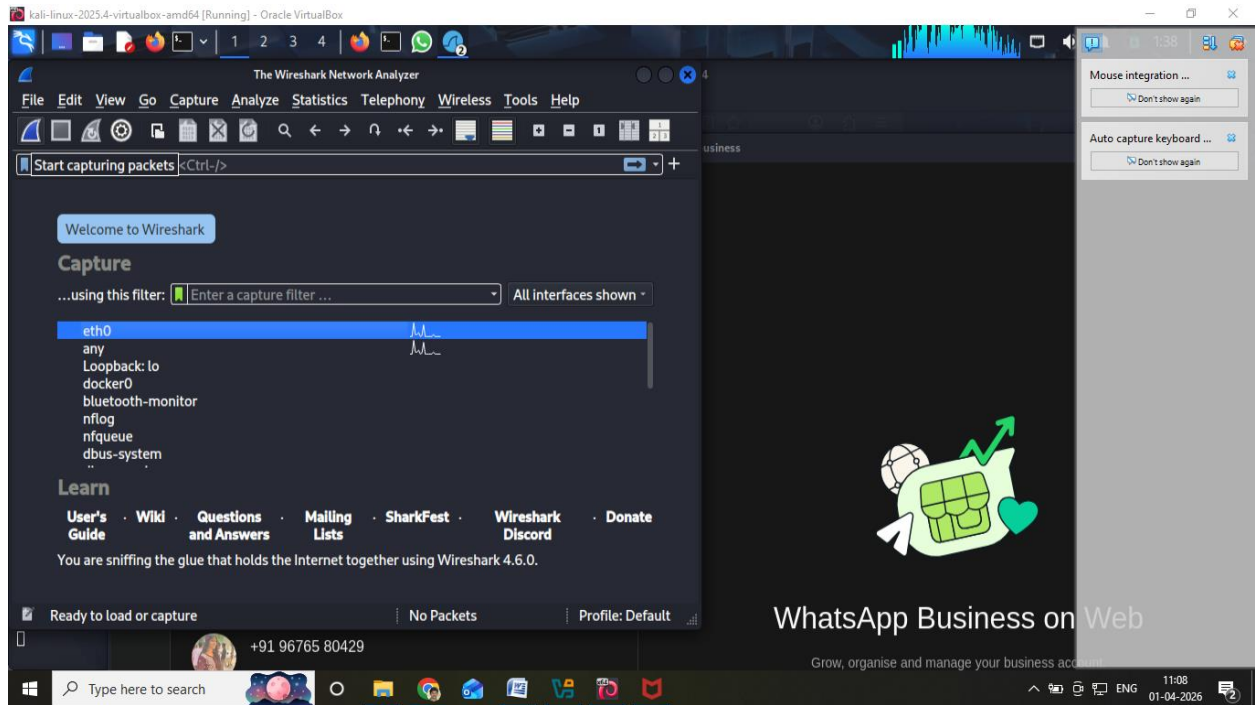
- Number of trackers
- Permissions used

Step 2: Start Wireshark

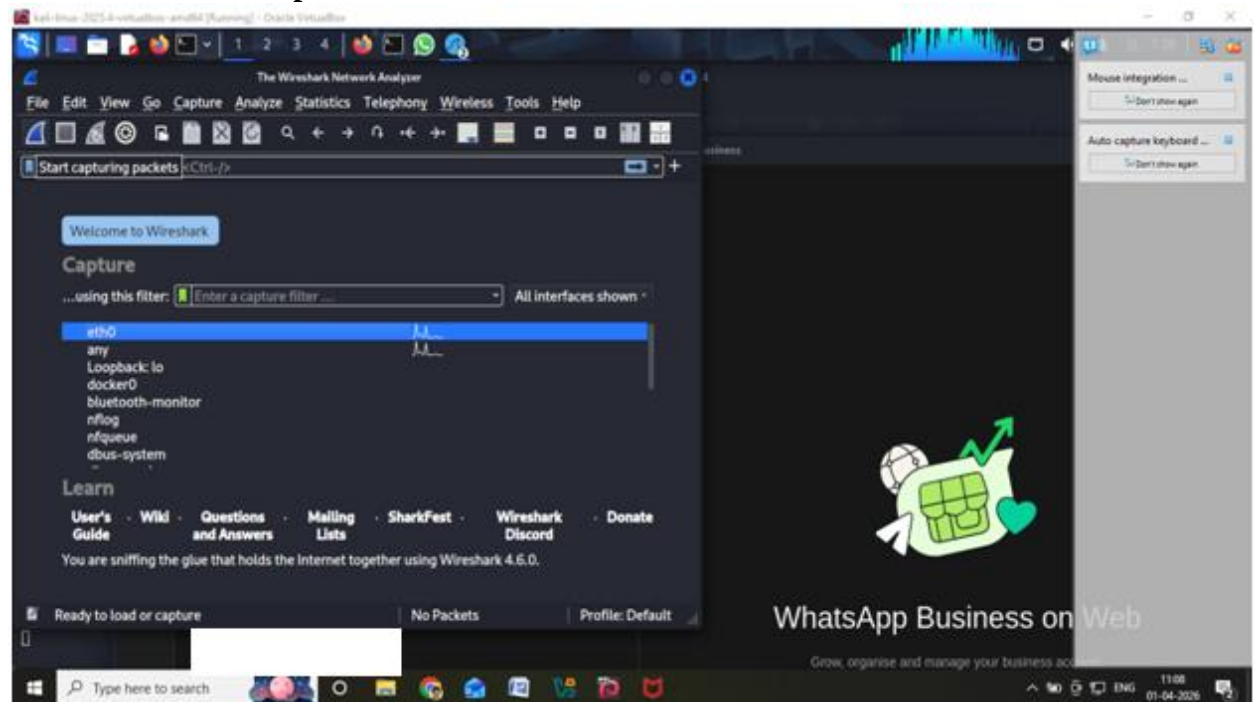
Wireshark



- Select interface (eth0 / wlan0)



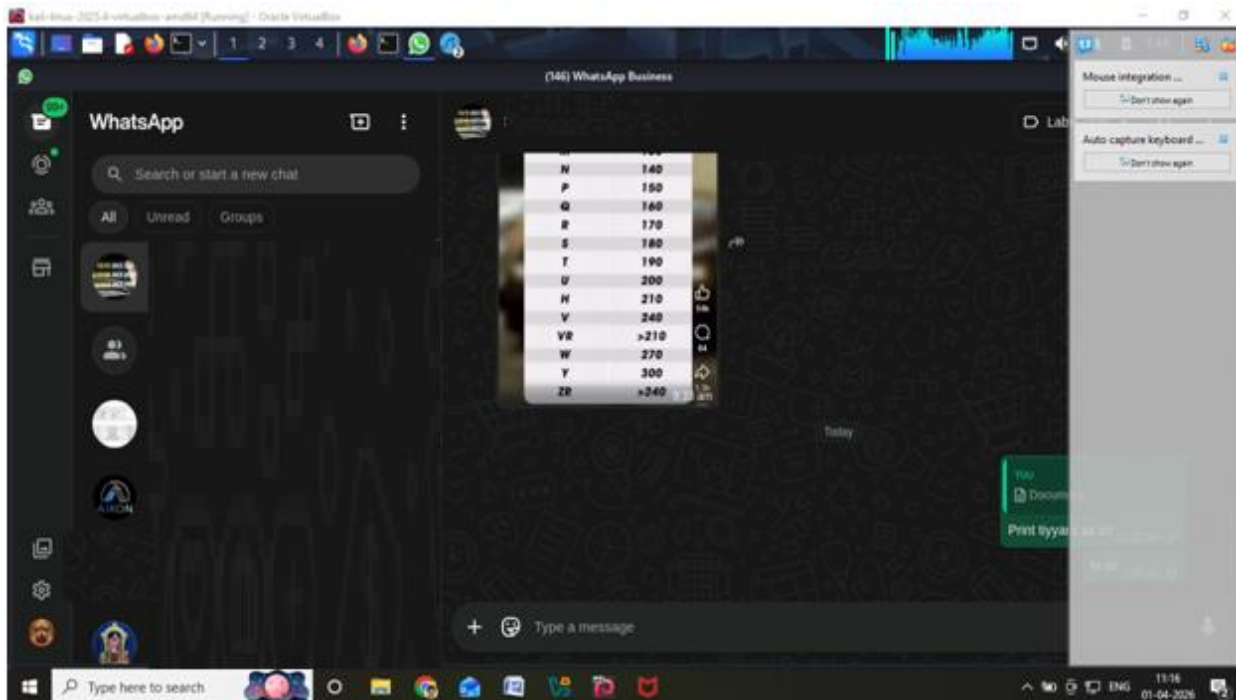
-
- Click Start Capture



Step 3: Generate Traffic

In WhatsApp Desktop:

- Send messages
- Send images
- Open chats

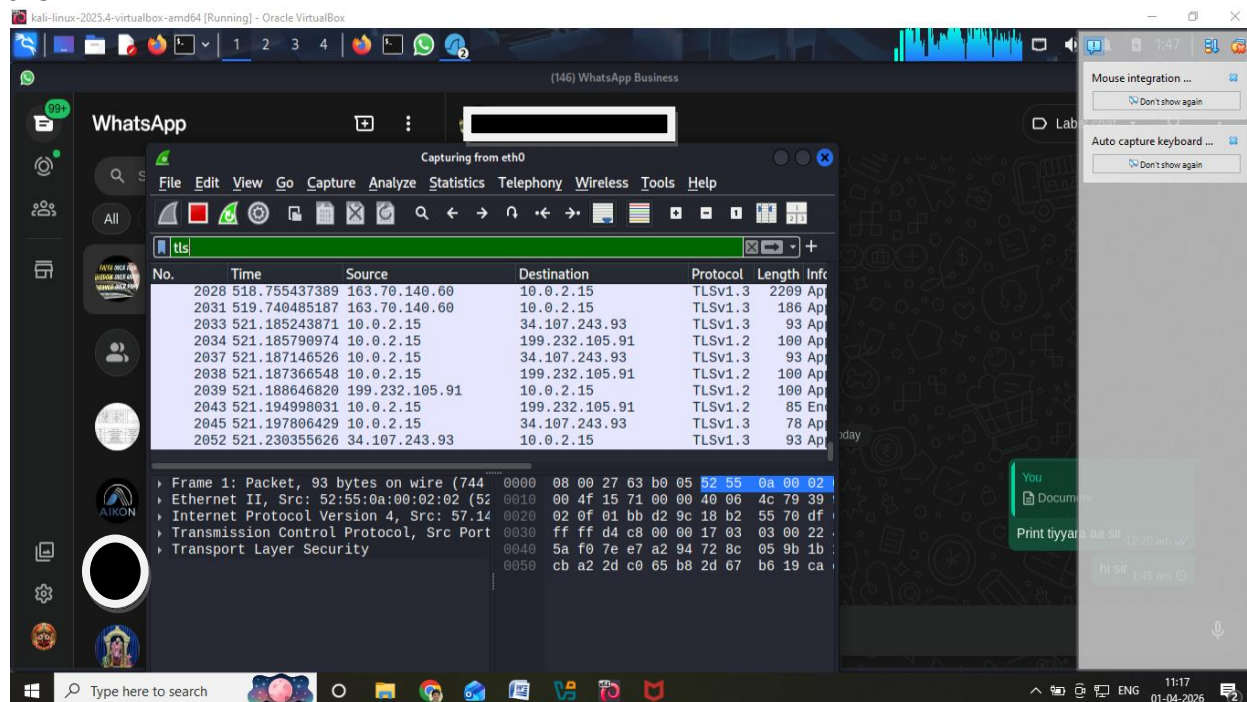


✓Step 4: Apply Filters

Use filters:

tls

dns

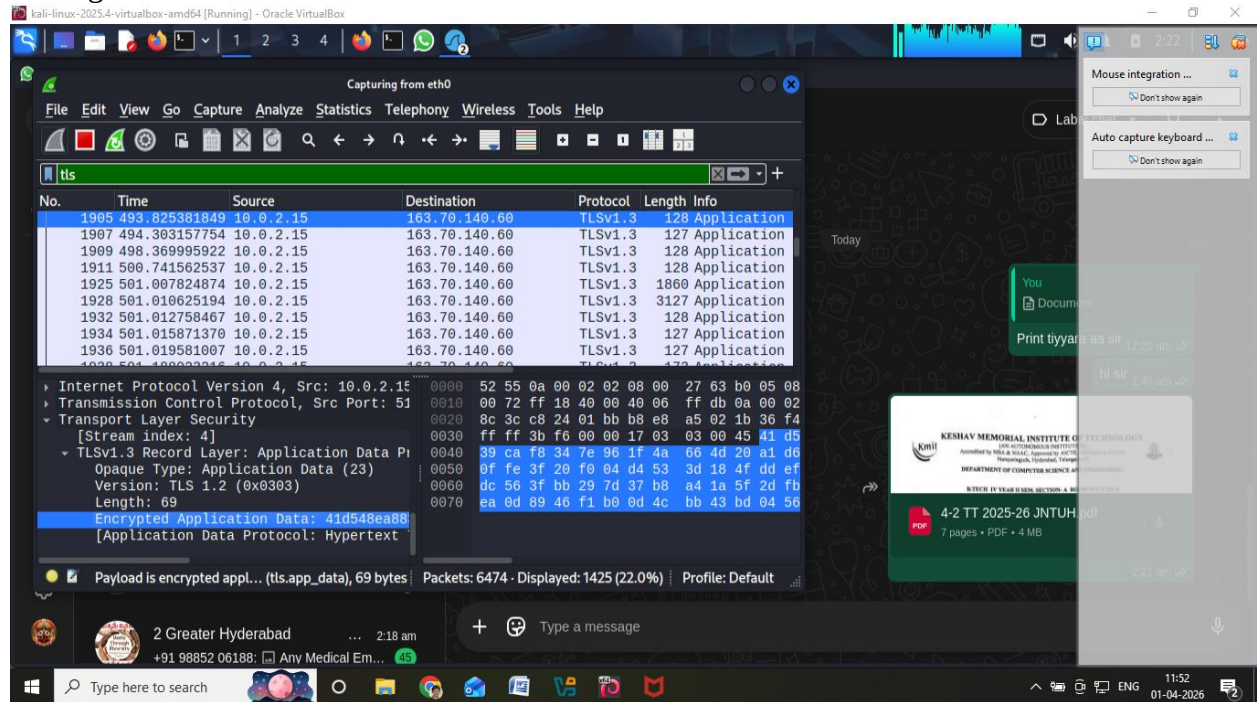


Observations

- Encrypted traffic (TLS packets)
- Domains:

- whatsapp.net
- facebook.com

Message content NOT visible



Findings

End-to-End Encryption

Secure communication

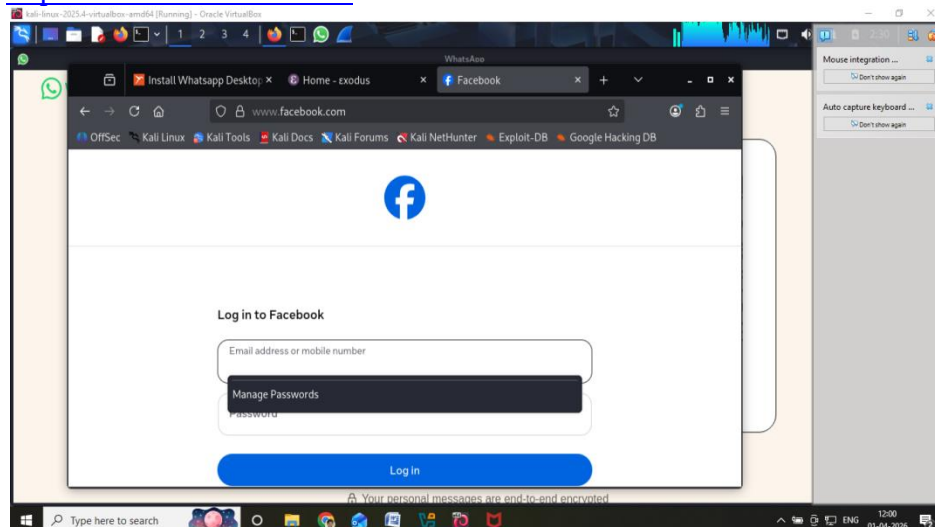
Metadata visible:

- IP address
- Time
- Server communication

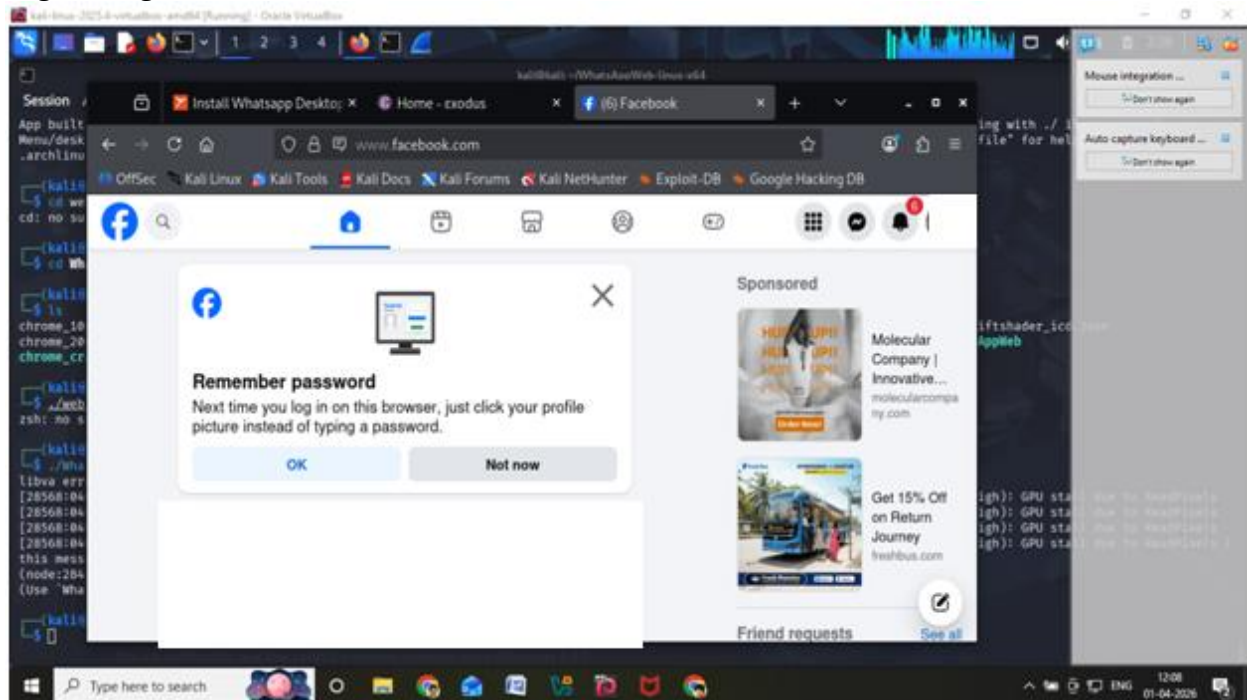
PART 2: Privacy Audit of Facebook Website

Step 1: Open Facebook

<https://www.facebook.com>

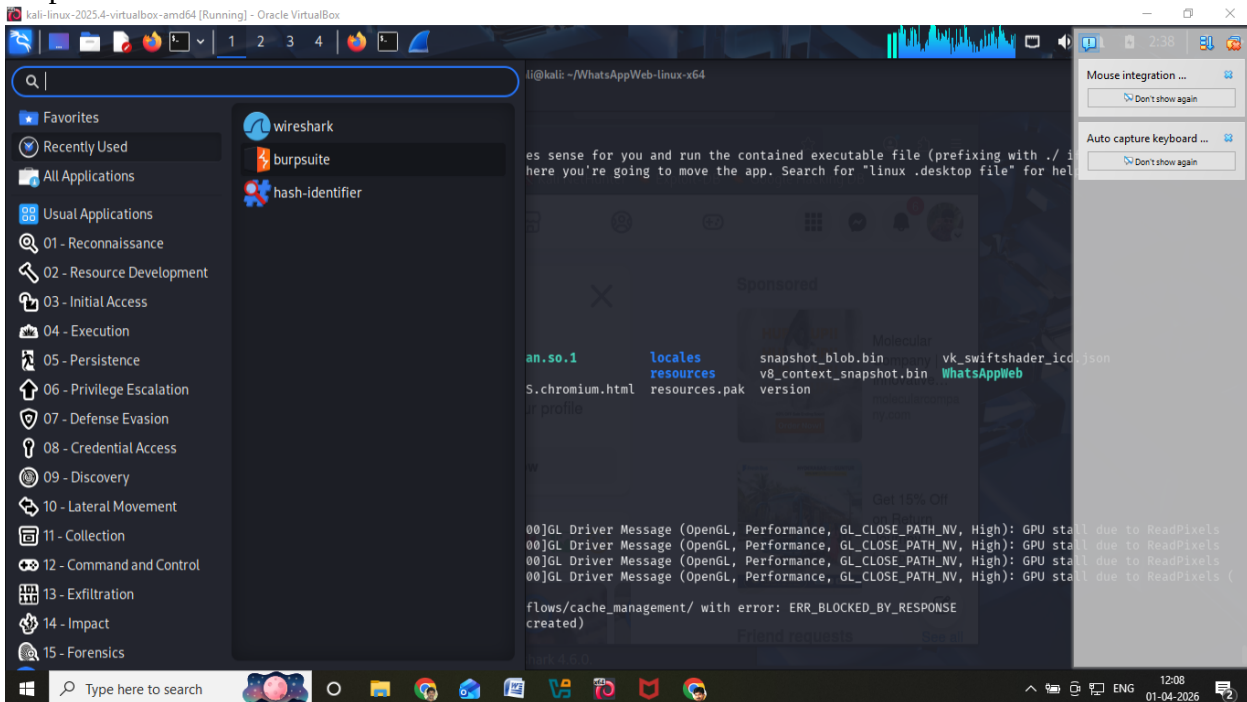


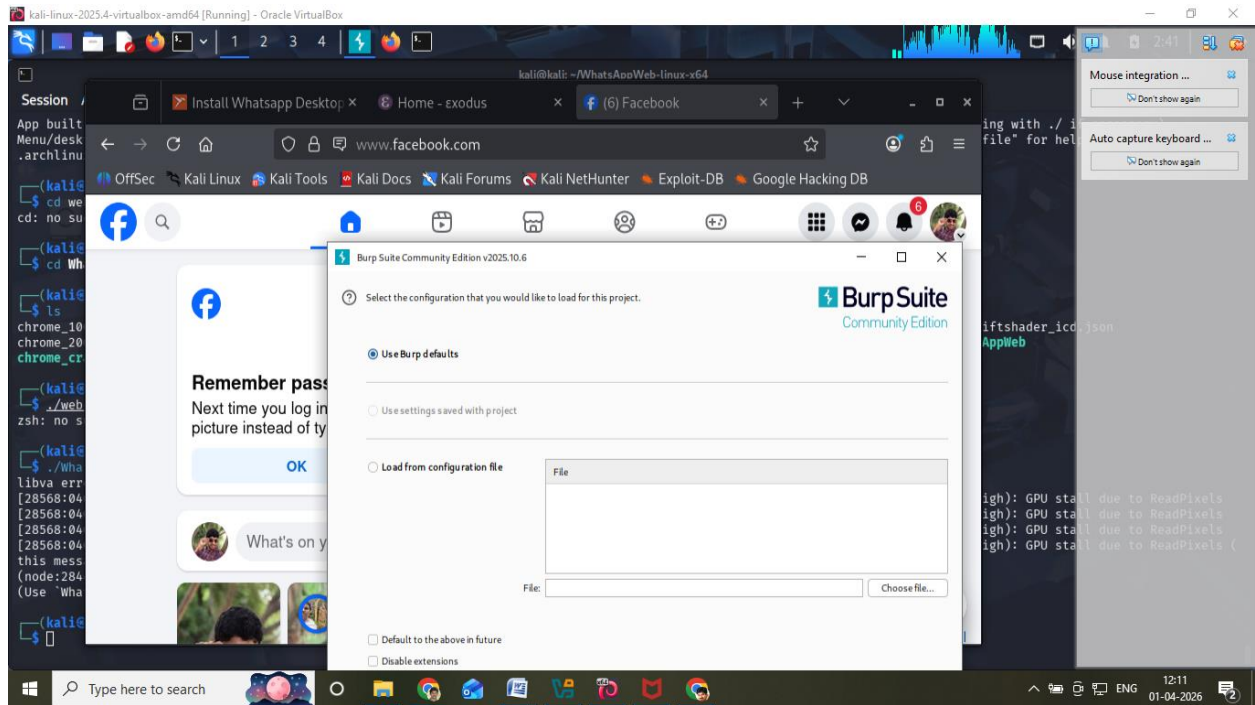
Login using test account



Step 2: Start Burp Suite

Burpsuite



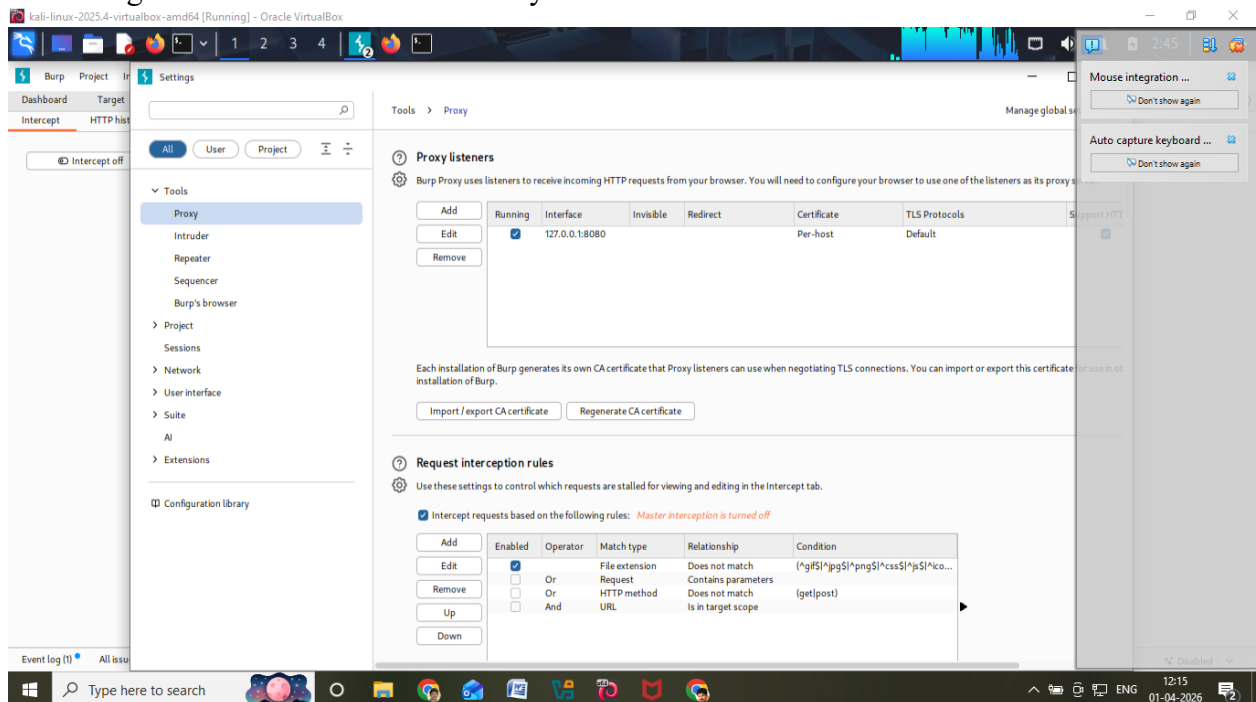


✓ Step 3: Configure Proxy

- IP: 127.0.0.1
- Port: 8080

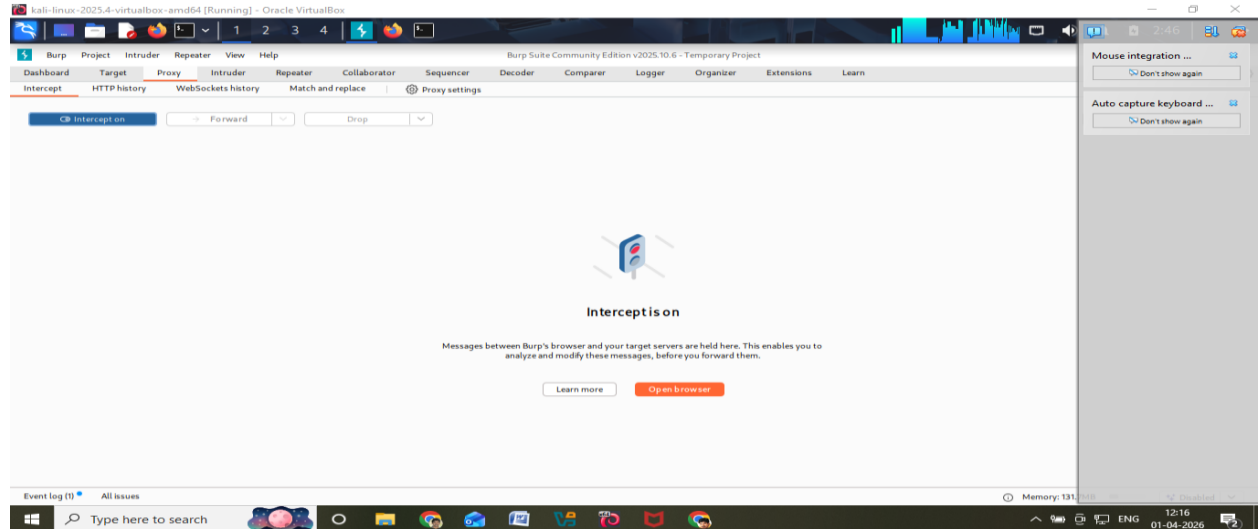
In browser:

- Settings → Network → Manual Proxy



✓ Step 4: Intercept Traffic

- Turn **Intercept ON**
- Reload Facebook



Observe:

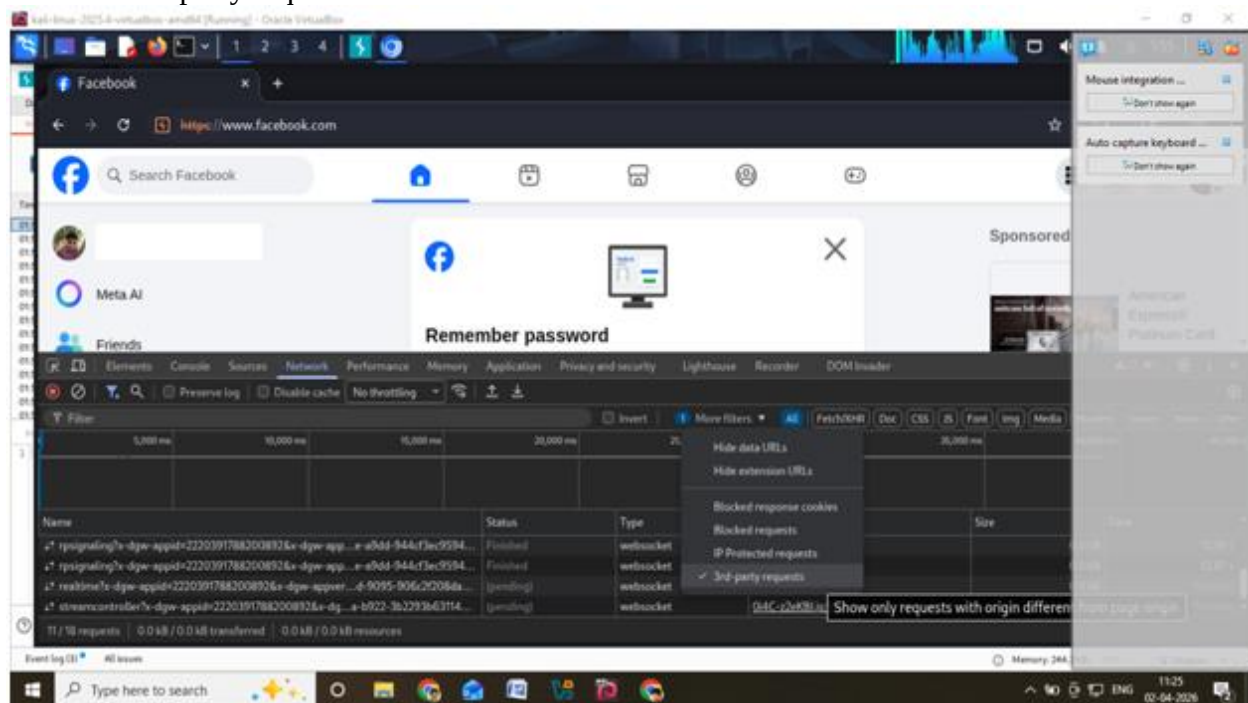
- Cookies (datr, fr)
- Session IDs
- Headers

Step 5: Analyze Tracking

Press **F12** → **Network** tab

Look for:

- Ads
- Analytics scripts
- Third-party requests



Findings

Heavy tracking

Uses cookies for personalization

Third-party integrations

PART 3: Data Breach Case Study

Case Study 1: Facebook Data Breach

- 533 million users affected
- Data leaked:
 - Phone numbers
 - Emails

Cause:

- API vulnerability

Case Study 2: WhatsApp Pegasus Attack

- Spyware attack via missed call
- Remote device access

Step 3: Check Email Breach

Go to:

□ <https://haveibeenpwned.com>

Enter email → check breach status

Analysis Points

- Type of attack
- Vulnerability
- Data exposed
- Prevention

Observations

- WhatsApp uses strong encryption
- Facebook performs extensive tracking
- Data breaches occur due to vulnerabilities

Result

Privacy audit was successfully performed using Kali Linux tools. WhatsApp ensures secure communication through encryption, while Facebook shows extensive tracking behavior. Data breach analysis highlighted risks of poor security practices.